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Viscous effects on the added mass and damping forces during free heave decay of a floating cylinder with a hemispherical bottom

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ABSTRACT

This paper presents an analysis of viscous effects on the added mass and damping forces during free heave decay of a cylinder with a hemispherical bottom. Results from unsteady Reynolds-averaged Navier–Stokes (URANS) simulations and linear potential-flow solutions are compared. Expressions for the cycle-averaged viscous added mass and damping coefficients are derived based on first-order Fourier analysis of the URANS results. Viscous effects are extracted by comparing the URANS results and linear potential-flow solutions, as nonlinear potential-flow effects in this case are demonstrated to be negligible. It is observed that viscous effects on the added mass coefficient are minor during heave decay, such that the linear potential-flow model can predict the value of the added mass coefficient to within 5% deviations. In contrast, the viscous damping coefficient exhibits strong cycle-dependent variations. A parametric study of the effects of initial displacement, cylinder diameter and draft shows that viscous damping forces can play an important role when the cylinder has a relatively small diameter and large draft. Further, the damping coefficient is transformed to the nondimensional drag force coefficient, and its dependence on the Reynolds number Re is investigated, for Keulegan–Carpenter numbers in the range $KC = 1.5$–2.5. It is found that the drag force coefficient is likely to be unrelated to Re and KC during the first cycle. From the second cycle to the fourth cycle, the drag force coefficient rapidly decreases with increasing Re, up to $Re \approx 3 \times 10^5$.

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1. Introduction

Floating cylinders are fundamental components in man-made floating offshore structures [1]. They are widely used in semi-submersible platforms, buoys for oceanographic research and point-absorbing wave energy converters [2]. Interactions between incident waves and the structures affect the safety and performance of the structures, with the added mass and damping forces being key parameters [3]. Potential-flow theory has been widely used for computation of added mass and damping forces [4–9]. However, viscous effects can be important for many cylindrical offshore structures, e.g., wave energy devices [10–14], offshore oil and gas platform [15–17] and floating offshore wind turbines [18–20]. In viscous fluids, there can be local separations and vortex shedding, and the added mass and damping forces become interdependent and time-dependent [21]. This motivates the present paper to investigate viscous effects on the added mass and damping coefficients.

Some existing studies address the oscillating viscous flow around a horizontal cylinder, or oscillating horizontal cylinder in immersed viscous fluid without a free surface. Stokes [22] was among the first to show the force on an oscillating body in viscous fluid is a linear combination of the added mass and damping forces, when the flow can be assumed as unseparated and laminar. Therefore, the damping force is dominated by viscous shear stress and proportional to the linear velocity. The Stokes solution [22] was extended to high-frequency oscillatory viscous flows and damped harmonic motion [23,24]. However, when the effects of flow separation and vortex shedding become important, the damping force deviates from the Stokes solution [22], as indicated by [25–27]. The oscillatory flow over a smooth circular cylinder was modelled at high Reynolds number assuming attached turbulent boundary layers [28]. The model gave

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Nomenclature

δ	Relative damping to the inertia
η	Heave displacement of the cylinder
η_0	Initial heave displacement
g	Gravity acceleration
ω	Oscillating frequency
ω_0	Natural oscillating frequency
ρ	Fluid density
ξ_{vis}	Nondimensional drag force coefficient
A	Water plane area under equilibrium state
a	Added mass coefficient for heave
b	Damping coefficient for heave
D	Diameter of the cylinder
d	Draft of the cylinder
k_0	Wave number
N	Number of oscillating cycle
T	Oscillation period
V	Displaced volume

good agreement with experimental results up to the point when separation occurs. Furthermore, the force on a circular cylinder in uniform, circular, orbital flows was investigated [29]. It was found that the viscous force due to steady streaming acts in opposition to the potential-flow force, which causes a substantial reduction in total force.

For floating structures oscillating in the vicinity of the free surface, the total damping force consists of a radiation damping force due to the free surface effect and a viscous damping force. Numerical experiments were conducted for a horizontal cylinder oscillating at the free surface at different frequencies [30]. It was observed that the added mass and damping coefficients are modified due to viscous effects, which is most prominent in the low-frequency range. Physical experiments were conducted for a horizontal cylinder submerged slightly beneath the free surface, and the measured load was observed to be significantly reduced compared to the load predicted by potential-flow theory [31]. The reduction is due to circulation generated around the cylinder by steady streaming, which is associated with viscous effects. For vertical cylinders, the added mass and damping coefficients for both surge and heave motion were investigated numerically [32]. It was found that viscous effects are stronger for surge motion than heave motion. With regard to different shapes, a cylinder with a hemispherical base produces a smaller viscous heave damping than the cylinder with the same diameter but a flat base. Viscous effects on the hydrodynamic performance and power conversion efficiency were also investigated numerically and experimentally for scaled point-absorber wave energy converters [33,34]. Viscous effects are non-negligible close to resonance, and must be taken into account in order to achieve a reasonable energy conversion efficiency [35], especially when wave gets large [36]. The hydrodynamic responses of the three-float wave energy converters M4 were investigated both numerically and experimentally in a series of publications [37–39]. The results from the linear time-domain potential-flow solver were compared with the experimental data, and viscous effects on the results were discussed.

A plausible way to account for viscous effects on wave loads for floating structures is to apply an ad-hoc drag force (proportional to velocity squared) to potential-flow theory, which makes the problem is quasi-steady. The magnitude of the drag force coefficient may differ from the one that is appropriate for steady flow at comparable Reynolds number, and has to be calibrated on a case-by-case basis. However, many wave–structure interaction problems are transient in nature, and the coefficients should depend on the current flow conditions and past flow history in addition to cylinder shape [40]. Nevertheless, constant coefficients are still used in the majority of studies [41]. This indicates that understanding of viscous effects is still limited, and further analysis is needed to bridge the gap between the behaviour of viscous flow and selection of drag force coefficients.

In the present paper, investigations with numerical and analytical approaches are conducted to address the viscous effects on the added mass and damping forces for free heave decay of a cylinder. Free decay is transient, and the body velocity varies significantly within a few heave cycles before the energy of the cylinder is expended. The free decay motion itself is simple enough that extraction of the linearised viscous added mass and damping coefficients is possible. We explore how the viscous forces evolve and deviate from the linear potential-flow solution.

The remainder of the paper is organised as follows. In Section 2, a boundary element method (BEM) based linear potential-flow solver and an unsteady Reynolds-averaged Navier–Stokes (URANS) solver are introduced, and expressions to extract the linearised added mass and damping coefficients from the URANS solutions are derived. Section 3 presents the results and discussions based on analysis of the numerical solutions. The primary focus is on the evolution of the viscous added mass and damping forces, and their magnitudes under different flow conditions. A parametric study is carried out, which covers the parameter space of initial displacement, diameter and draft. Finally, a summary is given in Section 4.

2. Numerical models

A cross-section of the problem is sketched in Fig. 1. A circular vertical cylinder with a hemispherical bottom is initially elevated above its equilibrium position at a height η_0. The cylinder has nominal diameter $D = 0.5$ m, draft $d = 0.5$ m (at equilibrium), and mass 81.8 kg. The diameter and draft are subjected to changes in Section 3 to investigate their effects on the added mass and damping forces. The water depth is $h = 3.5\,\text{m} \gg d$, i.e. the tests are conducted under deep-water conditions. The cylinder is released with zero velocity and undergoes free heave motion due to the initial vertical displacement. The other degrees of freedom are fixed. The cylinder oscillations create waves that radiate away from the cylinder.

In order to quantify viscous effects on the added mass and damping forces, both a URANS model and a linear potential-flow model are employed to compute the free decay motion for the above described cylinder. The added mass and radiation damping coefficients can be extracted directly from the potential-flow model introduced in Section 2.1. In contrast, a Fourier analysis approach is used to extract the added mass and damping coefficients from the URANS solution in Section 2.2. In Section 2.1, we demonstrate that the nonlinear potential-flow effects are negligible. Hence the deviation between URANS and linear potential-flow solution is mainly due to viscous effects.

2.1. Linear potential-flow model

An in-house code is used to solve the linear potential-flow model in the frequency domain, which consists of Laplace's equation and the linearised free surface and body boundary conditions. A boundary element method is used to solve the frequency-domain problem, in which a 3D free surface Green's function

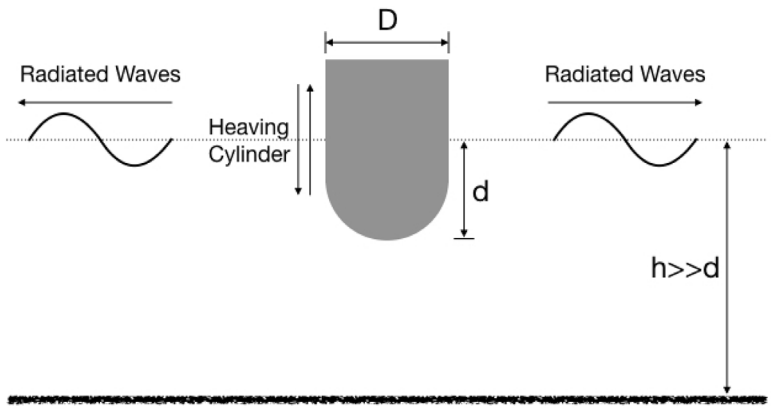


Fig. 1. Definition sketch of the problem: A cylinder with hemispherical bottom undergoing free heave due to an initial displacement η_0, where D is the cylinder diameter, d is the draft of the cylinder at equilibrium state and h is the water depth. The problem is in deep-water conditions, i.e. $h \gg d$.

Table 1
Geometric and hydrodynamic properties of the cylinder under the nominal condition.

Item	D [m]	d [m]	V [m^3]	a [–]	b [Ns/m]	ω_0 [rad]	δ [m^2/s^3]
Value	0.5	0.5	0.0818	0.21	18.90	4.42	0.095

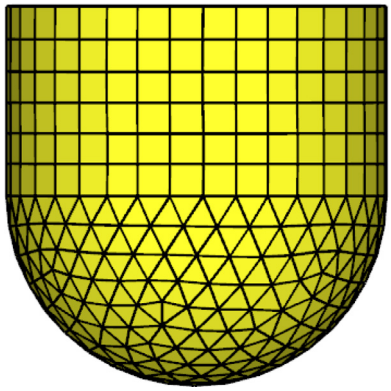


Fig. 2. Example of mesh used in the linear potential model. The upper cylindrical structure was meshed by quadrilateral elements while the surface in the lower hemisphere part was divided into triangular panels.

is employed and singularities with unknown strengths are distributed only on the body surface [42]. An example of the mesh used in the linear potential-flow model is shown in Fig. 2, where the body surface of the cylinder was divided into a number of quadrilateral and triangular panels. The geometric and hydrodynamic parameters from the linear potential-flow model are summarised in Table 1 for the cylinder oscillating at natural frequency.

As shown in Table 1, the added mass coefficient, a, and damping coefficient, b, are directly extracted from the linear potential-flow model. Since the body motion is assumed small and the instantaneous body surface is linearised, a and b only depend on the body shape and oscillating frequency. The free decay motion of the cylinder is approximated as (see Appendix A)

$$\eta(t) = \eta_0 e^{-\delta t}\cos(\omega_0 t) \quad \Rightarrow \quad \dot{\eta}(t) = -\eta_0\omega_0 e^{-\delta t}\sin(\omega_0 t) \quad \text{and}$$
$$\ddot{\eta}(t) = -\eta_0\omega_0^2 e^{-\delta t}\cos(\omega_0 t), \tag{1}$$

where ω_0 is the natural oscillation frequency and the damping rate is

$$\delta = \frac{b}{2\rho V(1+a)}. \tag{2}$$

The above approximation of the full linear solution in the time domain is based on the solution from linear potential-flow model in the frequency domain. In order to demonstrate that the nonlinear potential-flow effects are minor, we compare with the results from the nonlinear potential-flow solver ANSYS-AQWA in Fig. 3, in which the equation of motion is solved in time domain and the nonlinear free-surface condition is applied. Hence, it is solved on the moving surface and the nonlinear pressure is applied on the instantaneous wetted surface [43]. The displacements given by ANSYS-AQWA and normalised by the initial displacements collapse with one another, indicating negligible influence of the nonlinear terms in the potential-flow model. Moreover, Eqs. (1) agrees with the nonlinear potential-flow solutions, providing evidence that it is an accurate approximation. Therefore, viscous effects are considered to be the dominant source of any deviations of the added mass and damping coefficients between those extracted from the URANS solution and those from Eq. (1).

2.2. URANS model

An URANS model is set up using the OpenFOAM software package [44]. It consists of the incompressible continuity and Navier–Stokes momentum equations for a two-phase flow system of water and air in an Eulerian frame. The free surface is captured by the volume of fluid (VoF) method [45], and a k–ω SST turbulence model is applied in the numerical model, which models the turbulent kinetic energy and the specific dissipation rate. The model produces a time series of fully nonlinear forces on the cylinder, which include both the pressure and the shear stress induced forces. The heave acceleration is obtained by solving the equation of motion for the cylinder in the vertical direction.

The computational model is set up in 2D, based on the axisymmetric geometry and the Reynolds-averaged flow. The length and height of the numerical tank were chosen as 5 m and 3.5 m, respectively. This ensures that the tank bottom has negligible effects on the decay of the cylinder, and the reflected waves do not affect the cylinder in the first four oscillating periods. Moreover, active wave absorption boundaries were applied at the outlet boundaries to prevent reflection of the radiated waves [46].

An overview of the mesh structure is presented in Fig. 4. In the vertical direction, the cells in the base mesh are squeezed

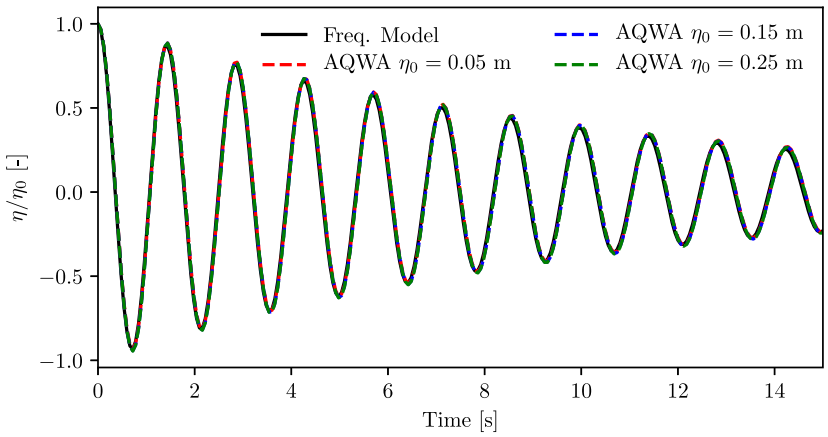


Fig. 3. Comparison of the free heave decay motion from the time-domain nonlinear potential-flow solver AQWA-NAUT and the frequency-domain linear potential-flow solver.

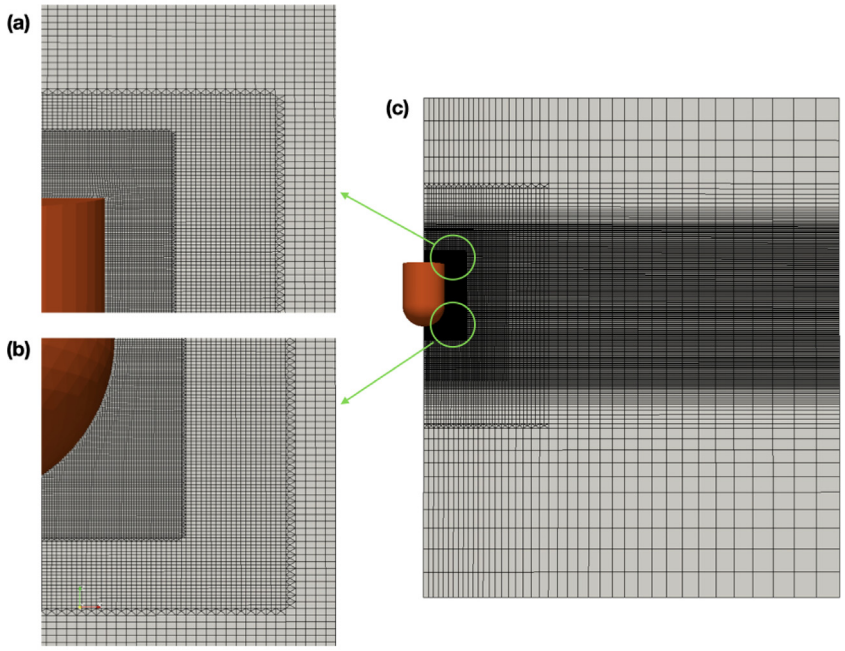


Fig. 4. Overview of the mesh structure used in the RANS model. The cylinder is highlighted in ochre.

around the free surface, and coarsened in the atmosphere and the bottom boundary (Fig. 4c). In the horizontal direction, they are squeezed near the cylinder and uniformly distributed within that area. The base mesh is refined by four levels horizontally and five levels vertically, which ensures that the cell aspect ratio is nearly unity and the resolution near the cylinder is sufficiently high. The geometry of the cylinder was represented by a stereo lithography surface file. The base mesh gradually conforms to the surface by iteratively refining the base mesh and morphing the resulting split-hex mesh to the surface. Convergence and validation tests for this set of mesh are carried out to determine the mesh resolution that is used in the simulations and validate the URANS against experimental data. Readers are referred to Appendix B for details.

2.3. Added mass and damping coefficients from URANS solution

The added mass and damping coefficients are not explicitly given in the URANS solution. Recalling that the added mass is the force component in phase with the acceleration, i.e., $\sin(\omega t)$, and the damping is the force in phase with velocity, i.e. $\cos(\omega t)$, they are extracted using first-order Fourier analysis. The time series of the vertical force is expanded as

$$F_3(t) = a_0 + \sum_{n=1}^{P}(a_n \cos(n\omega t) + b_n \sin(n\omega t)), \tag{3}$$

where the coefficients are

$$\begin{aligned} a_0 &= \frac{1}{NT}\int_0^{NT} F_3(t)\,\mathrm{d}t, \\ a_n &= \frac{2}{NT}\int_0^{NT} F_3(t)\cos(n\omega t)\,\mathrm{d}t, \\ b_n &= \frac{2}{NT}\int_0^{NT} F_3(t)\sin(n\omega t)\,\mathrm{d}t, \end{aligned} \tag{4}$$

in which N is the number of cycles to be used in the Fourier analysis, and T is the oscillating period.

Based on linear potential-flow theory, the vertical dynamic force on the cylinder, F_3, can be approximated in terms of the

added mass force, the radiation damping force, and the restoring force, as

$$F_3 = -\rho V a\ddot{\eta} - b\dot{\eta} - \rho g A\eta. \tag{5}$$

By substituting Eq. (5) into Eq. (4), the added mass and damping coefficients are, respectively,

$$a = -\frac{\int_0^{NT}(F + \rho g A\eta)\cos(\omega t)\,\mathrm{d}t}{\rho V \int_0^{NT}\ddot{\eta}\cos(\omega t)\,\mathrm{d}t}, \tag{6}$$

$$b = -\frac{\int_0^{NT}(F + \rho g A\eta)\sin(\omega t)\,\mathrm{d}t}{\int_0^{NT}\dot{\eta}\sin(\omega t)\,\mathrm{d}t}. \tag{7}$$

The heave decay expression, i.e. Eq. (1), is substituted into (6) and (7) to obtain

$$a = \frac{2\delta\int_0^{NT}(F + \rho g A)\cos(\omega t)\,\mathrm{d}t}{\omega^2 V\rho\eta^0(1 - e^{-NT\delta})}, \tag{8}$$

$$b = \frac{2\delta\int_0^{NT}(F + \rho g A\eta)\sin(\omega t)\,\mathrm{d}t}{\omega\eta^0(1 - e^{-NT\delta})}, \tag{9}$$

where the following approximations have been used

$$\int_0^{NT}\dot{\eta}\sin(\omega t)\,\mathrm{d}t = -\frac{2\eta_0\omega^3}{\delta(\delta^2 + 4\omega^2)}(1 - e^{-NT\delta}) \approx -\frac{\omega\eta_0}{2\delta}(1 - e^{-NT\delta}), \tag{10}$$

$$\int_0^{NT}\ddot{\eta}\cos(\omega t)\,\mathrm{d}t = \frac{\eta_0\omega^2}{\delta}\left(\frac{\delta^2 + 2\omega^2}{\delta^2 + 4\omega^2}\right)(1 - e^{-NT\delta}) \approx \frac{\eta_0\omega^2}{2\delta}(1 - e^{-NT\delta}). \tag{11}$$

This set of formulations are validated in Appendix C, where it is shown that Eqs. (8)–(9) are sufficiently accurate to extract the added mass and damping coefficients from the URANS solutions.

3. Results and discussion

3.1. Evolution of the viscous added mass and damping coefficients

We present an overview of the evolution of the added mass and damping coefficients extracted from the URANS solution (a_{URANS} and b_{URANS}, respectively) for the case with initial displacement $\eta_0 = 0.20$ m. Fig. 5 shows the coefficients normalised with respect to the corresponding coefficients from the linear potential-flow model, a_{BEM} and b_{BEM}, which are constants. The normalised added mass coefficient decreases over time, while the damping coefficient increases. At the first cycle, $N = 1$, the URANS added mass coefficient is 3.9% larger than the linear potential-flow prediction. Over the subsequent two cycles, the URANS added mass coefficient approaches the linear potential-flow solution, and reaches a steady value, 3.0% greater than linear potential-flow solution at the third and fourth cycles.

The URANS damping coefficient is ≈1% larger than the linear potential-flow prediction at the first cycle. The similarity is attributed to the damping force being dominated by radiation of waves just after it is released, with the small deviation likely due to the shear stress induced force. Fig. 6 presents the vorticity field near the cylinder at six different times. At the beginning of the first cycle, $t = 0.1$ s, a thin boundary layer is formed around the cylinder. The boundary layer is thicker at $t = 0.5\,\mathrm{s} = 0.36T$, which is close to the trough of the first cylinder cycle. When the cylinder moves up from the trough near to the peak at $t = 1.3\,\mathrm{s} = 0.94T$, minor detachment of boundary layer is observed below the hemispherical base of the cylinder. The bottom row of panels in Fig. 6 show that the flow further develops during the subsequent cycles, over $t = 1.9$–2.8 s, or $t = 1.38T$–$2.03T$. The resulting viscous effects are reflected in the damping coefficient in Fig. 5, which move further away from the linear potential-flow prediction as the number of cycles increases. The deviation is particularly large between the first and second cycles, with the difference increasing up to 13.5% at $N = 2$. At the third and fourth cycles, the damping coefficient is a steady 26% greater than the linear potential-flow prediction.

3.2. Effects of initial displacement, cylinder draft and diameter

Analysis of URANS solutions shows the variations of the added mass and damping coefficients with respect to the number of cycles are consistent with the findings described above in Section 3.1 for most cases. Therefore, in this section we focus on changes in the average amplitudes of the added mass and damping coefficients over the first four cycles against those caused by changes in the initial displacement η_0, cylinder draft d and diameter D.

Fig. 7 shows deviations of the added mass and damping coefficients from the linear potential-flow predictions against initial displacement. The deviations in the added mass and damping coefficients increase with increasing initial displacement. This is expected, as linear potential-flow theory assumes small displacements, allowing the pressure to be integrated to the mean free surface rather than the instantaneous free surface. However, the deviations in the added mass coefficient are small, increasing from ≈3% at $\eta_0 = 0.05$ m only up to ≈ 4.75% at $\eta_0 = 0.25$ m. In contrast, viscous effects on the damping coefficient are non-negligible even when the initial displacement is small, e.g., the damping coefficient from URANS solution is 15% larger than that from the linear potential-flow solution for $\eta_0 = 0.05$ m. The deviations increase up to 23% at $\eta_0 = 0.25$ m.

Fig. 8 shows the deviations in the added mass and damping coefficients against cylinder diameter. The deviations decrease with increasing diameter, indicating viscous effects decrease with increasing diameter. This suggests that the inertia effects become more important when the diameter of the cylinder increases. As above, the deviations in the added mass coefficient are small, reducing from 7% at $D = 0.35$ m to 2% at $D = 0.6$ m. The deviations in the damping coefficient drop significantly from greater than 50% at $D = 0.35$ m to less than 10% at $D = 0.6$ m. The considerable deviation in the damping coefficient for the smallest diameter highlights the importance of viscous damping in the computation of wave–structure interaction problems.

Fig. 9 shows the deviations against draft. The deviations in the added mass coefficient increase from ≈1.5% at $d = 0.35$ m to ≈5% at $d = 0.6$ m. The deviations in the damping coefficient are between 7.5%–10% for $d = 0.35$–0.45 m, and then increase to 25% at $d = 0.6$ m. This indicates viscous effects become more important as draft increases, and shows linear potential-flow theory can be inaccurate for cylinders with large drafts.

3.3. Comments on the drag force coefficient

As discussed in Section 1, an adhoc quadratic drag force term (proportional to velocity squared) is usually used to describe the viscous effects on the damping force

$$F_{vis} = -\frac{1}{2}\zeta_{vis}\rho A\dot{\eta}|\dot{\eta}|, \tag{12}$$

where $\zeta_{vis}(t)$ is the drag force coefficient, which can be related to a linear viscous damping coefficient $b_{vis} = b_{URANS} - b_{BEM}$. The relationship is based on the principle that an identical amount of

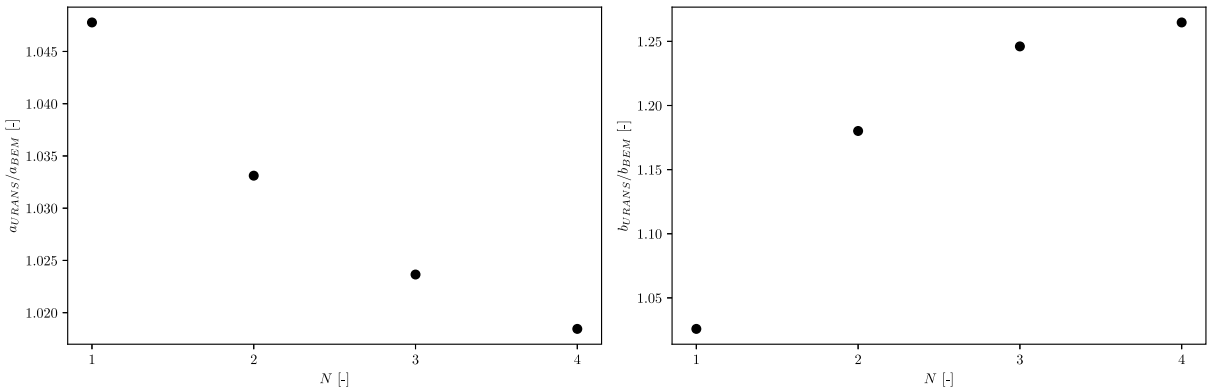


Fig. 5. Variation of the relative added mass coefficient a_{URANS}/a_{BEM} (left) and relative damping coefficient b_{URANS}/b_{BEM} (right) during the first N cycles of the free heave decay. a_{URANS} and b_{URANS} are the added mass and damping coefficients obtained from the URANS solution. a_{BEM} and b_{BEM} are the added mass and damping coefficients obtained from the linear potential-flow solution. The initial displacement is $\eta_0 = 0.2$ m.

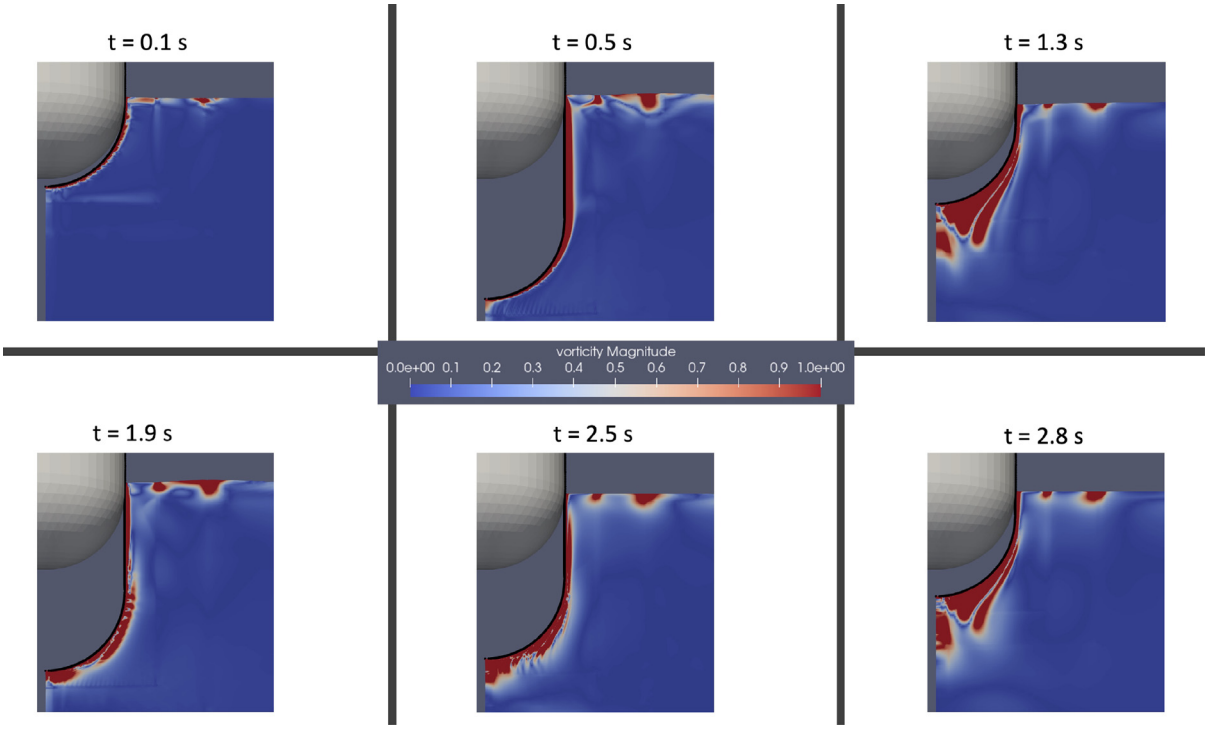


Fig. 6. Example of the vorticity field near the cylinder at different time instants for the case with $\eta_0 = 0.2$ m. The solid cylinder represents the initial position of the cylinder at $t = 0$, and the black curve represents the instantaneous boundary of the cylinder.

energy is dissipated by a linear term with an equivalent linear damping coefficient during the decay [47], such that

$$\int_0^{NT} -b_{vis}(\dot{\eta})^2 \, \mathrm{d}t = \int_0^{NT} -\frac{1}{2}\zeta_{vis}\rho A(\dot{\eta})^2|\dot{\eta}| \, \mathrm{d}t. \tag{13}$$

This results in the equivalent linearised viscous damping term

$$b_{vis} = \frac{3}{16}\pi\rho A\zeta_{vis}\dot{\eta}^{rms}, \tag{14}$$

where $\dot{\eta}^{rms}$ is the root mean square of $\dot{\eta}$, i.e.

$$\dot{\eta}^{rms} = \sqrt{\frac{1}{NT}\int_0^{NT}[\dot{\eta}(t)]^2 \, \mathrm{d}t} = \frac{\eta_0\omega}{2}\sqrt{\frac{1-e^{-2NT\delta}}{NT\delta}}. \tag{15}$$

Hence, the nondimensional drag force coefficient ζ_{vis} yields

$$\zeta_{vis} = \frac{32b_{vis}}{3\pi\rho A\eta_0\omega}\sqrt{\frac{NT\delta}{1-e^{-2NT\delta}}}. \tag{16}$$

The drag force coefficient on a cylinder immersed in quasi-steady oscillatory flow is dependent on the Keulegan–Carpenter number KC and the Reynolds number Re [1]. Therefore, when considering the viscous effects on the damping force as shown in Eq. (12), it is commonly assumed that the drag force coefficient ζ_{vis} for the heaving cylinder is also a function of KC and Re. For free heave decay, we define KC and Re as

$$Re = \dot{\eta}_{max}D/\nu \quad \text{and} \quad KC = \dot{\eta}_{max}T/D, \tag{17}$$

where $\dot{\eta}_{max}$ is the maximum velocity of the cylinder in heave during a given cycle.

In Fig. 10, ζ_{vis} is plotted as a function of Re, for KC in the range 0.5–1.5. It incorporates all the data points in Figs. 7–9. During the first cycle, $\zeta_{vis} \approx 0$ for most cases, and does not appear to be related to Re. This is consistent with the finding in Section 3.1 that the viscous damping at the first cycle is negligible, and the damping force is dominated by wave radiation. From the second cycle, ζ_{vis} decreases from ≈0.2 at $Re \approx 5 \times 10^4$ to ≈0.05 at $Re \approx$

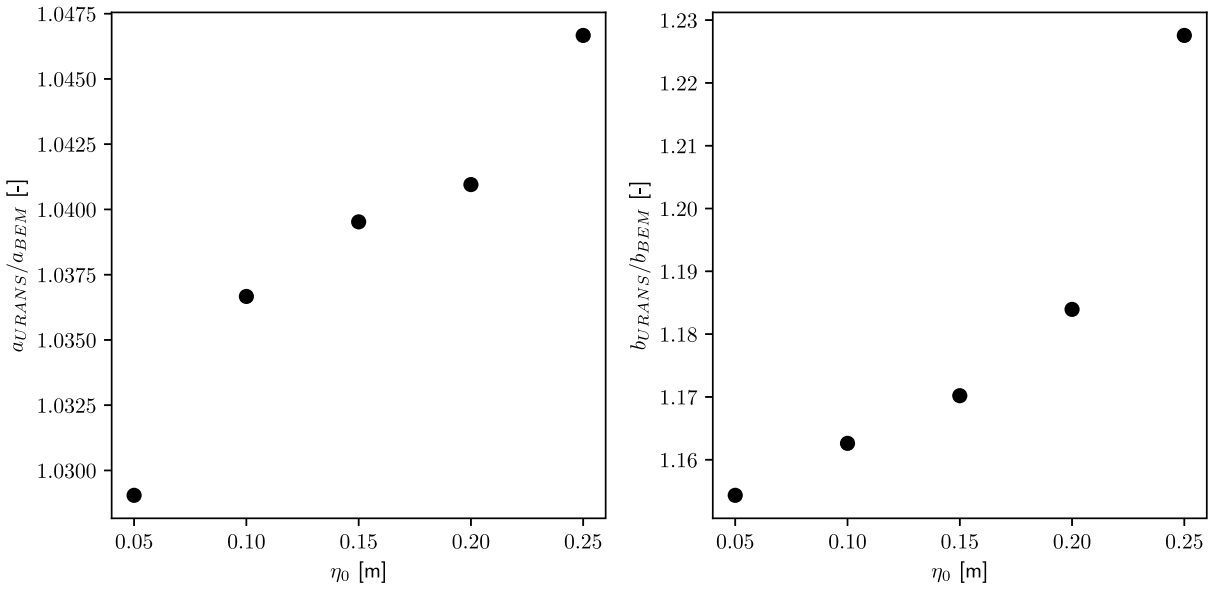


Fig. 7. Comparison of the relative deviation of the added mass coefficient (left) and the damping coefficient (right) between the URANS solution and the linear potential solution under different initial displacements η_0. The diameter of the cylinder is $D = 0.5$ m and the draft of the cylinder is $d = 0.5$ m.

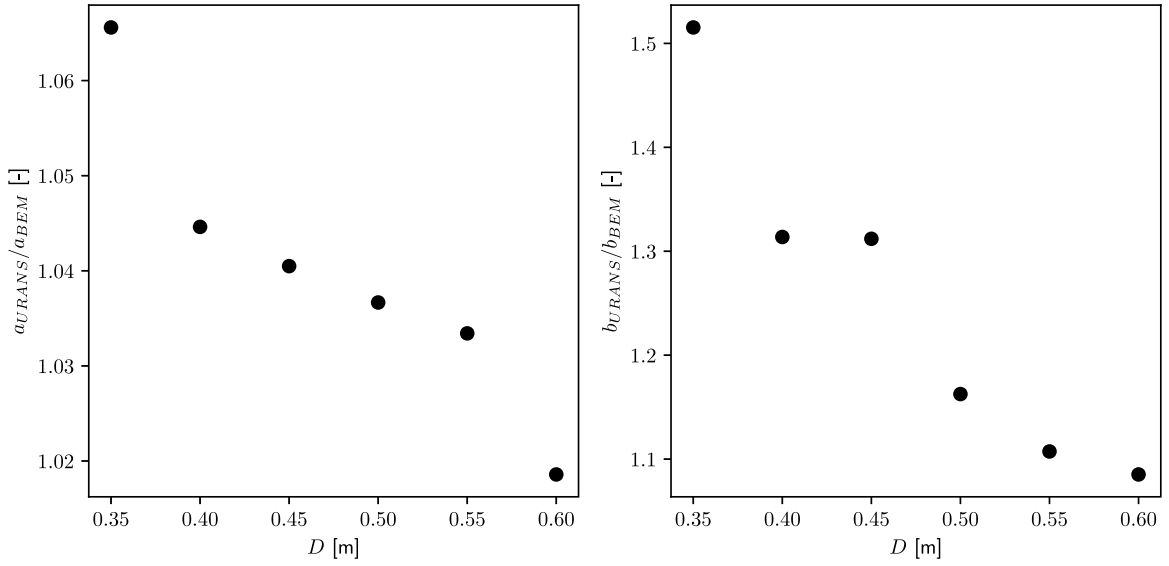


Fig. 8. Comparison of the relative discrepancy of the added mass coefficient and the damping coefficient between the URANS solution and the linear potential solution under different diameters D. The draft is $d = 0.5$ m and the initial displacement is $\eta_0 = 0.10$ m.

3×10^5. The data points fit well into an exponentially decaying curve $\zeta_{vis} = 0.33 \exp(-7.023 \times 10^{-6} Re)$, and most of the data points fall within the ±30% error band, as shown in Fig. 10. The probable reasons for the (moderate) scatter of the data points are that: (1) The drag force coefficient is KC dependent, i.e. although the range of KC for the data points in Fig. 10 is restricted to 0.5–1.5, its effect is not fully eliminated, which causes scatter of the data points. However, the scatter is unlikely to reduce even if KC range is further reduced, as (2) the flow for free heave decay of the cylinder is not a quasi-steady oscillatory flow. Thus, the drag force coefficient may be affected by the transient effects and not solely dependent on KC and Re, so that variations of ζ_{vis} can be anticipated even for the same Re and KC.

4. Conclusions

A series of free decay tests for a cylinder with hemispherical bottom have been conducted using an URANS model and a linear potential-flow model. The added mass and damping coefficients were extracted from both models, and viscous effects on these coefficients were analysed.

Analysis of the evolution of the added mass coefficient from the URANS solution showed the coefficient decreases over time and approaches the linear potential-flow solution. The deviation of the added mass coefficient between the URANS solution and the linear-potential flow solution was shown to be small in general, indicating that viscous effects are minor for the added mass coefficient. The damping coefficient was shown to exhibit a rapid increase during the heave decay of the cylinder. The damping coefficient from the URANS solution was shown to be close to the linear potential-flow solution at the first cycle, but then rapidly grow in the following cycles and deviate considerably from the linear potential-flow solution. This demonstrates the importance of viscous effects on the damping coefficient.

A parametric study was conducted to examine the effects of draft, diameter and initial displacement on the added mass and

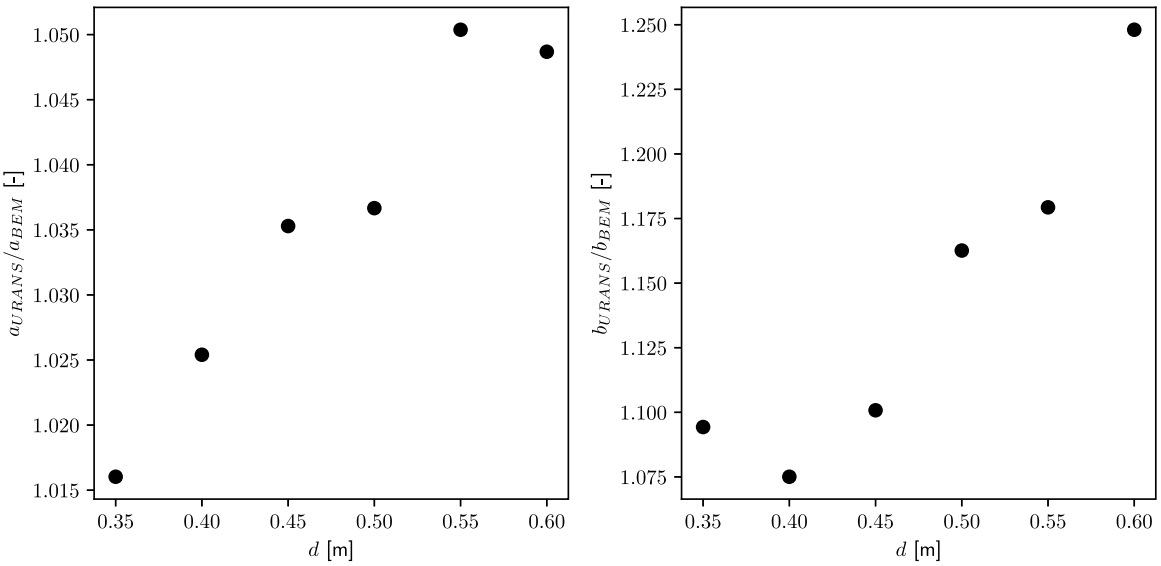


Fig. 9. Comparison of the relative deviation of the added mass coefficient and the damping coefficient between the URANS solution and the linear potential solution under different drafts d. The diameter is $D = 0.5$ m and the initial displacement is $\eta_0 = 0.10$ m.

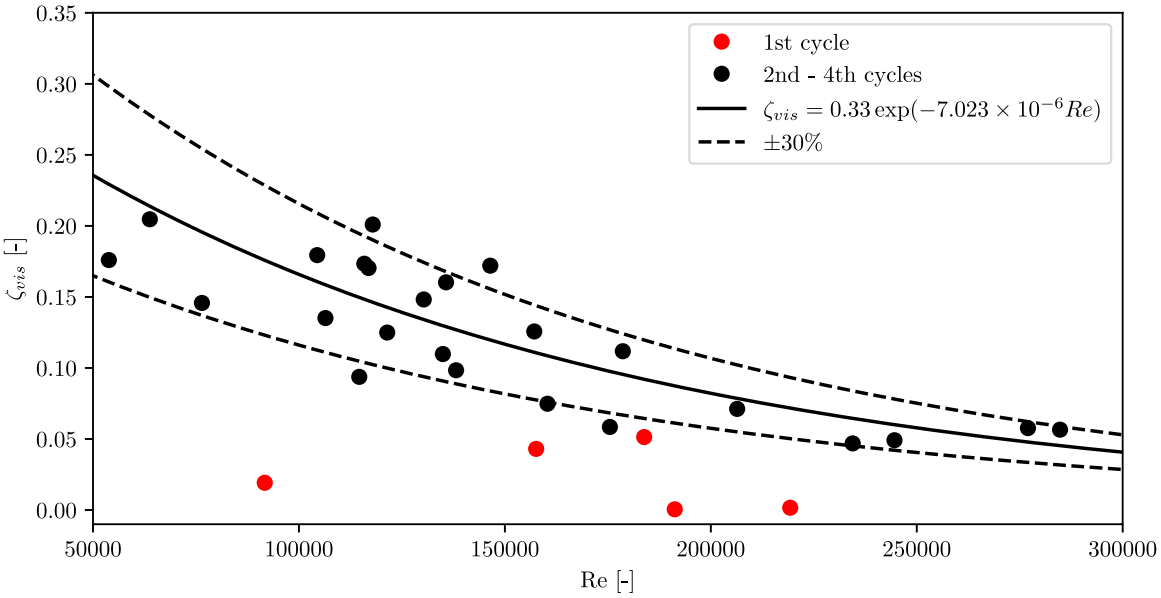


Fig. 10. The drag force coefficient ζ_{vis} against Re. $KC = 0.5$–1.5.

damping coefficients. Viscous effects on the added mass coefficients are limited, and the linear potential-flow model predictions are accurate to within 10%. In contrast, the draft, diameter and initial displacement have a major effect on the viscous damping coefficient. The viscous damping force was shown to be most important when the diameter of the cylinder is small and the initial displacement and the draft of the cylinder are large.

The linear viscous damping coefficient was transformed to a quadratic drag force coefficient. The drag force coefficient was found to be unrelated to Re during the first cycle, as the viscous effects are generally small. During subsequent cycles, the drag force coefficient decreases with increasing Re, up to $Re \approx 3 \times 10^5$ with $KC = 0.5$–1.5. The data points were fit into an exponentially decaying curve with an $\pm 30\%$ error band.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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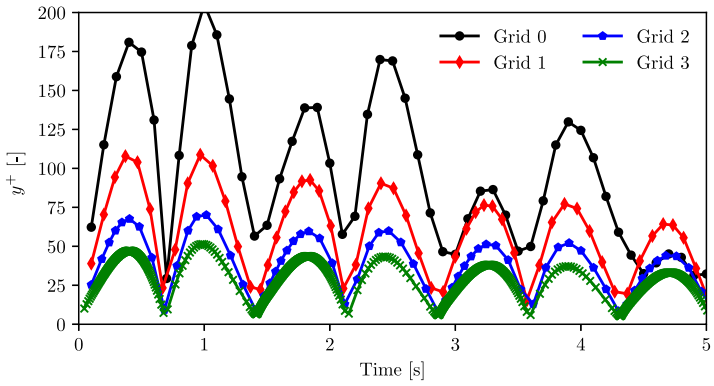


Fig. B.11. Time history of the instantaneous spatially averaged y^+ value at the cylinder boundary with different grid resolutions.

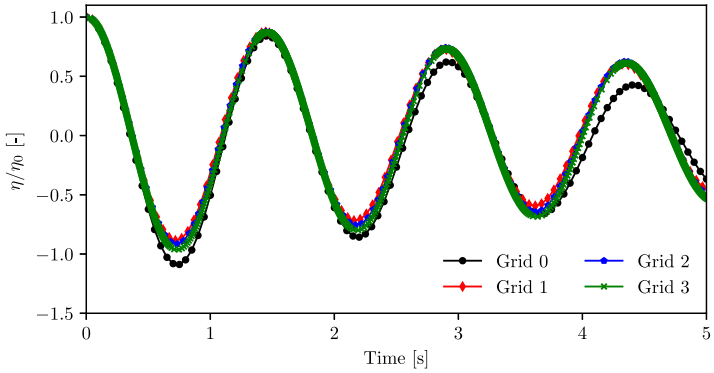


Fig. B.12. Time history of heave decay of the cylinder for grids of different resolution. Relevant mesh parameters for heave decay are given in Table B.2.

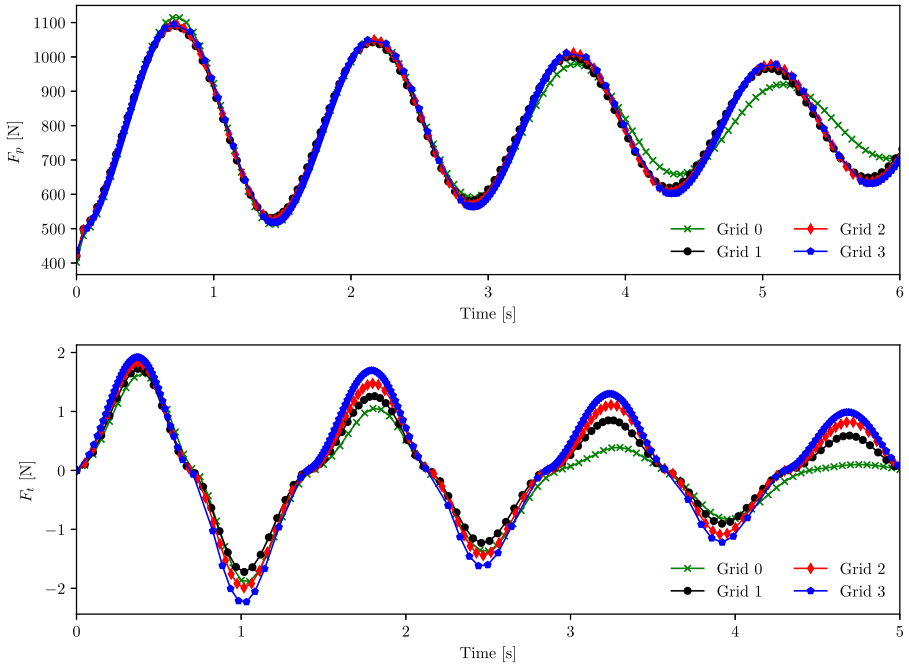


Fig. B.13. The integrated force on the cylinder with different grid resolutions. Upper: the integrated pressure force F_p. Lower: the integrated shear stress force F_t. The relevant mesh parameters for the heave decay is shown in Table B.2.

Appendix A. Linear heave decay equation

In this section, we derive an approximate linear heave decay equation for the cylinder. For a cylinder heaving at the free surface, the dynamic vertical force acting on the cylinder can be categorised as the added mass force, damping force and restoring force, so that

$$F_3 = -\rho V a\ddot{\eta} - b\dot{\eta} - \rho g A\eta. \tag{A.1}$$

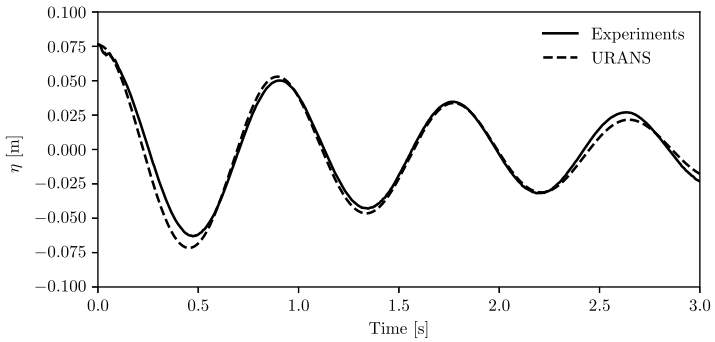


Fig. B.14. Validations of the URANS solutions on heave decay against the experimental data in [32].

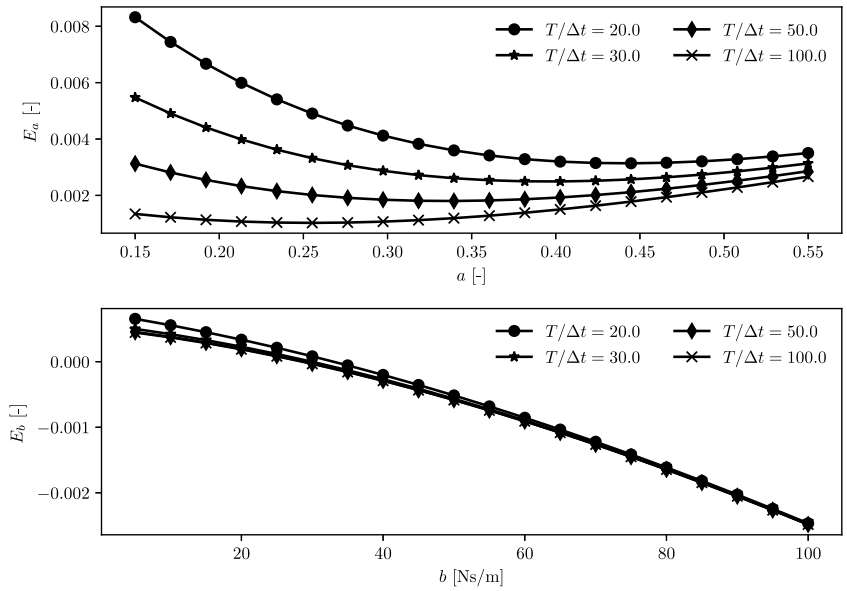


Fig. C.15. Relative error in added mass coefficient (upper) and relative error in damping coefficient (lower) for the first set of verification tests. The provided force signals were the perfect added mass and damping force based on the linear potential model, respectively. ΔT is the resolution of the force signal. T is the oscillating period.

The added mass coefficient a and the damping coefficient b are assumed to be constant, with values taken at the resonant frequency. By applying the momentum conservation law, the motion equation of the cylinder reads

$$(1+a)\rho V\ddot{\eta} + b\dot{\eta} + \rho g A\eta = 0. \tag{A.2}$$

With the initial conditions $\eta|_{t=0} = \eta_0$ and $\dot{\eta}|_{t=0} = 0$, the solution to Eq. (A.2) is

$$\eta = \eta_0 e^{-\delta t}\left(\cos(\omega_z t) + \frac{\delta}{\omega_z}\sin(\omega_z t)\right), \tag{A.3}$$

where ω_z is the damped oscillating circular frequency and δ is a damping coefficient relative to the inertia of the system

$$\delta = \frac{b}{2\rho V(1+a)}. \tag{A.4}$$

The damped frequency is related to the undamped frequency by

$$\omega_z^2 = \omega_0^2 - \delta^2, \tag{A.5}$$

where the undamped frequency is expressed as

$$\omega_0 = \sqrt{\frac{gA}{V(1+a)}}. \tag{A.6}$$

The damping of the cylinder is generally small compared to the inertia ($\delta < 0.2$). Hence, $\delta \ll \omega$ holds true and no explicit differentiation is given between the damped and undamped frequency

$$\omega_z = \omega_0. \tag{A.7}$$

Moreover, by applying the condition $\delta \ll \omega$, Eq. (A.3) is further simplified to

$$\eta = \eta_0 e^{-\delta t}\cos(\omega_0 t). \tag{A.8}$$

The velocity of cylinder is obtained by taking the derivative of the decay with respect to time

$$\dot{\eta} = \frac{\partial \eta}{\partial t} = -\eta_0 e^{-\delta t}\sqrt{(\delta^2+\omega_0^2)}\sin(\omega_0 t + \phi), \tag{A.9}$$

where ϕ is the phase difference between the velocity and the displacement

$$\phi = \arccos\left(\frac{\omega_0}{\sqrt{\omega_0^2+\delta^2}}\right) \approx 0. \tag{A.10}$$

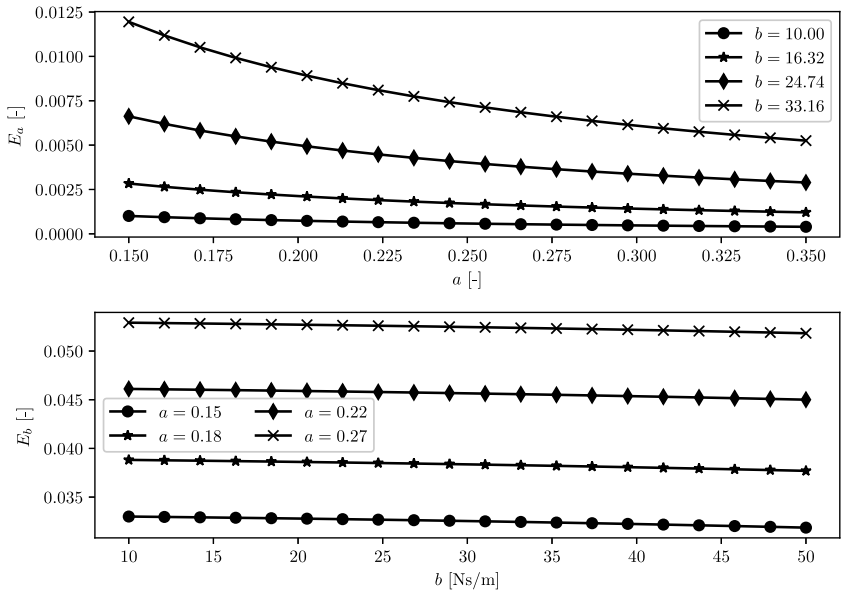


Fig. C.16. Relative error in added mass coefficient (upper) and relative error in damping coefficient (lower) for the second set of verification tests. The provided force signals were the summation of the added mass and damping forces based on the linear potential model.

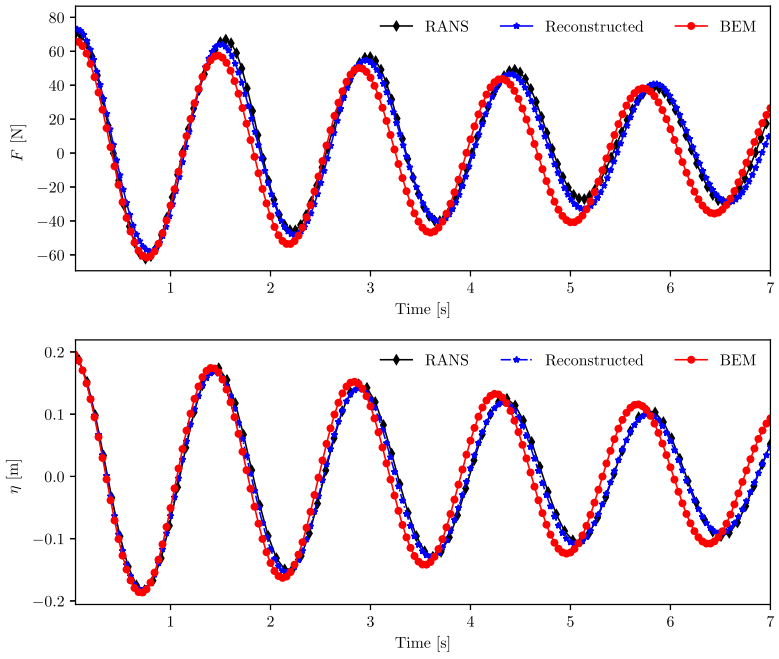


Fig. C.17. Comparison of the force (upper) and heave motion (lower) between the RANS model, the linear potential model and the reconstructed results. The reconstructed results utilised Eq. (5) with the coefficients extracted based on Eqs. (8)–(9). The initial displacement in this case is $\eta_0 = 0.2$ m.

The velocity of cylinder is simplified to

$$\dot{\eta} = -\eta_0 \omega_0 e^{-\delta t} \sin(\omega_0 t). \tag{A.11}$$

By similar manipulation, the acceleration yields

$$\ddot{\eta} = \frac{\partial^2 \eta}{\partial t^2} = -\eta_0 \omega_0^2 e^{-\delta t} \cos(\omega_0 t). \tag{A.12}$$

Appendix B. Convergence and validation for the URANS model

A convergence test was carried out for the URANS model in order to determine the mesh resolution that should be used in the simulations. Four different grid resolutions were tested with a refinement factor of two in each direction. The parameters related to grid resolutions are presented in Table B.2. The spatially averaged y^+ value on the cylinder boundary patch, representing the mesh resolution in the near wall region, is plotted against time in Fig. B.11. It is noticed that from Grid 1, the grid resolution is sufficient in principle to keep y^+ smaller than 100 during the decay process.

In the present work, the quantities of great interest are the decay of the cylinder, the integrated pressure force and shear stress force on the cylinder. The results of the convergence test on these quantities are presented in Fig. B.12 and Fig. B.13, respectively. In terms of heave decay, it exhibits a very good convergence property from Grid 1 while Grid 0 overestimated the heave decay period. This indicates that the added mass predicted by Grid 0 is higher than the other grids. Meanwhile, the trough

Table B.2
Mesh parameters for four different grid resolutions. Δx and Δz are the grid length and height near the cylinder.

Grid ID [–]	Δx [m]	Δz [m]	Mesh NO. [–]
0	0.01490	0.0104	2075
1	0.00745	0.0051	9100
2	0.00373	0.00255	37000
3	0.00186	0.00127	148000

of the decay in Grid 0 is also considerably larger. For instance, the first trough of the decay displacement differs from the others for at least 18%. From Grid 2, the numerical solution for the heave decay is sufficiently convergent, and the difference is less than 4%. Regarding the force, the pressure induced force, which is dominant in this case, exhibits very good convergence property from Grid 1. However, stress force tends to increase monotonically as the resolution increases, and an absolutely convergent solution is difficult to obtain even with the finest grid resolution. This appears the same as stated in Ref. [32], which concluded that numerical convergence is not achieved for the small shear stress. The difference between the finest two grids is about 8% for the first trough. In all the simulations in the present work, we decided to apply Grid 3, which offers a satisfactory compromise between accuracy and cost.

The URANS model is validated against a classical and widely used experimental data set from the laboratory tests in the Plymouth COAST basin [32]. A cylinder with a hemispherical base was used in the experiment with a diameter of 0.3 m and a mass of 9.75 kg. Under equilibrium state, the draft was 0.187 m. The initial elevation was η_0 =0.076 m. The displacement of the cylinder was obtained using the Qualisys motion capture system. The experiment was reproduced using the URANS numerical model, and the comparison of the cylinder displacement is presented in Fig. B.14. The trough at the first cycle is overestimated by the numerical model and the difference reaches 11%. The following cycles are predicted accurately by the model up to the peak at the third cycle, which has a deviation of 15%. In general, the decay period is predicted reasonably, but the decay speed of the cylinder in the numerical model is slightly faster than in the experiment.

Appendix C. Verification and validation of Eqs. (8)–(9)

Eqs. (8)–(9), which are proposed to extract the added mass and damping coefficients from the RANS solution for heave decay test of a cylinder, are carefully verified and validated in this section. Two sets of cases are conducted to verify Eqs. (8)–(9). In the first set of verification tests, the added mass and damping coefficients are evaluated from the following analytic perfect added mass and damping time series: $-\rho V a \eta_0 e^{-\delta t}\cos(\omega t)$ and $-b\eta_0\omega e^{-\delta t}\sin(\omega t)$. The main source of error arises from the resolution of the force signal, given that the signal is discrete and the integral in the numerator has to be evaluated numerically. The relative error between the extracted and the given coefficients is presented in Fig. C.15. It is found that the relative error is confined within about 0.1%. The added mass coefficient is more significantly affected by the resolution of the time series, but it has almost no effect on the damping coefficient. Actually, the sample frequency of the signal from the URANS solutions is much higher than the present ones, approximately in the level of $T/\Delta t = 10^3 \sim 10^4$. Therefore, the error due to sample frequency of the force signal is expected to be very small.

In the second set of verification tests, the force signals comprised a summation of added mass and damping force components given by $-\rho V a \eta_0 e^{-\delta t}\cos(\omega t) - b\eta_0\omega e^{-\delta t}\sin(\omega t)$. The test result is shown in Fig. C.16, where it is seen that the relative error is associated with the relative magnitude of these two force components. If the force signal is dominated by one component, the accuracy of the extracted force coefficient for that component is higher than for the other component. For instance, when extracting the added mass coefficient as shown in the upper figure of Fig. C.16, the lowest relative error occurs when the damping force coefficient is smallest, i.e. $b = 10$ Ns/m, as the added mass term weights more in the force signal. In the limit, where the added mass force is extremely large and completely dominates the overall force, the relative error of the added mass coefficient should be similar to the results in the first set of verification test. It is interesting that this approach always overpredicts the added mass and the damping coefficients. The deviation is generally smaller for the added mass force because the system is inertia dominant, as illustrated in Appendix A. Regarding the damping coefficient, the error increases when the added mass coefficient is increasing, which reaches up to 5%. This set of verification tests covers the parameter space encountered in the present work. Hence, we believe the error introduced in this approach should be within this range, i.e., less than 5%.

To further validate Eqs. (8)–(9), we utilised a force signal time series obtained from the URANS model as the input for Eqs. (8)–(9). The initial displacement $\eta_0 = 0.2$ m. We emphasise that this provided force signal is indeed fully nonlinear and may deviate from the analytical solution in Eq. (5). The extracted added mass and damping coefficients were then applied to reconstruct the force signal according to Eq. (5). Fig. C.17 compares the decay displacement and heave force time histories obtained using the URANS model, BEM model, and the reconstruction. The reconstructed signal provides a closer match to the URANS simulation because of its inherent influence on the added mass and damping coefficients.

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